Akam Nawzad Othman

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Education

University of Kurdistan Hewlêr (UKH) Erbil, Iraq

BSc in Software Engineering – Graduated top 3 in programme *Oct 2021 – Jul 2025*

- Thesis: *Fine-tuning XLM-RoBERTa for Named Entity Recognition in Kurmanji Kurdish*, published at WiNLP 2025, co-located with EMNLP 2025

AI Bootcamp – Certificate of Completion Erbil, Iraq

Ovanya *Nov 2024 – Dec 2024*

- Intensive programme on machine-learning fundamentals and neural networks

Experience

AI Developer Aug 2025 – Present

Ovanya *Erbil, Iraq*

- Developed OCR and document-processing pipelines for scanned Kurdish documents
- Contributed to building one of the first OCR systems and datasets for Sorani Kurdish
- Trained and evaluated machine-learning models for text extraction from scanned pages
- Built document classification and text segmentation workflows
- Deployed model inference services with **FastAPI**
- Developed evaluation benchmarks for extraction quality and model reliability

AI Researcher Sep 2024 – Oct 2024

UKH Artificial Intelligence and Innovation Centre *Erbil, Iraq*

- Collected and cleaned a **140K+** article Kurdish news corpus for NLP research
- Built scalable web-scraping pipelines with Python and Beautiful Soup
- Contributed to developing one of the first Kurdish spell-checking datasets
- Designed annotation guidelines and supported dataset labelling workflows
- Ran model evaluations and maintained dataset versioning for reproducible research

Publications

KuMMLU: A Multitask Language Understanding Benchmark for Central Kurdish 2026

- *Proceedings of the 10th Widening NLP Workshop (WiNLP 2026), co-located with EMNLP 2026*, Budapest, pp. 173–180 – paper | dataset
- 3,272 multiple-choice questions across 18 subjects: 2,256 native questions from national Grade 9 and Grade 12 exams, and 1,016 translated from English MMLU
- Evaluated 32 language models; the best reach about 88% accuracy, multilingual baselines score only modestly above chance, and every model does better on translated than on native questions
- Publicly released the benchmark and evaluation code

Fine-tuning XLM-RoBERTa for Named Entity Recognition in Kurmanji Kurdish 2025

- *Proceedings of the 9th Widening NLP Workshop (WiNLP 2025), co-located with EMNLP 2025* – paper
- Built a manually annotated Kurmanji NER dataset of 7,919 sentences (person, organisation, location)
- Fine-tuned XLM-RoBERTa to an F1 of 0.8735 (precision 0.8668, recall 0.8803); released the first public Kurmanji NER model with a web demo

Projects

Kurdish Text-to-Speech | *PyTorch, MMS, Hugging Face* Nov 2025

- Fine-tuned Meta's MMS TTS on a curated 10-hour Sorani dataset to produce the first open Sorani Kurdish TTS model; the demo supports Sorani and Badini
- Improved pronunciation clarity and naturalness through audio preprocessing, transcript normalisation and evaluation

Kurdish Spellchecker | *ByT5, Transformers, Gradio*

- Fine-tuned ByT5-base as a sequence-to-sequence corrector for misspelled Sorani sentences; byte-level modelling handles Kurdish character variation

KuOCR – Kurdish OCR and Document Extraction | *Computer Vision, FastAPI* Jun 2025 – Oct 2025

- Full OCR system for Sorani Kurdish: a large real and synthetic dataset, right-to-left text rendering with bounding boxes, and a layout-aware extraction pipeline served via FastAPI

Multilingual Legal RAG System | *FastAPI, Vector Search* Oct 2025 – Present

- Embedding-based search and a chatbot that answers questions over official legal documents using retrieved Kurdish, Arabic and English texts, with custom text preprocessing

Kurdish Handwritten Character Recognition | *Python, NumPy, Streamlit* May 2025 – Jun 2025

- Implemented a 547K-parameter CNN from scratch in NumPy (manual convolution, im2col, backpropagation); 71.05% accuracy across 35 classes

Gorrer and **KurdType** | *Python, Flask, Web*

- A Latin–Arabic Kurdish script transliterator, and a curated catalogue of open-licensed fonts that support Kurdish

Honours

Finalist, 12th Undergraduate Research & Innovation Competition (URIC 2025), Abu Dhabi May 2025

Technical Skills

AI/ML: NLP, Deep Learning (CNNs, Transformers), Computer Vision & OCR, Speech Synthesis, LLM Evaluation, RAG
Languages: Python, C++, SQL, JavaScript
Libraries: PyTorch, TensorFlow, Hugging Face, scikit-learn, NumPy, Pandas, NLTK, Matplotlib
Backend & Tools: FastAPI, Flask, PostgreSQL, MySQL, MongoDB, Vector Databases, Docker, Git, GitHub, GitLab
Spoken: Kurdish (native), English (C1), Arabic (B1–B2)